

Histogram

INFORMATIKA UNS – PENGOLAHAN CITRA DIGITAL GANJIL 2026/2027

Menghitung Histogram

Histogram

Menghitung Histogram

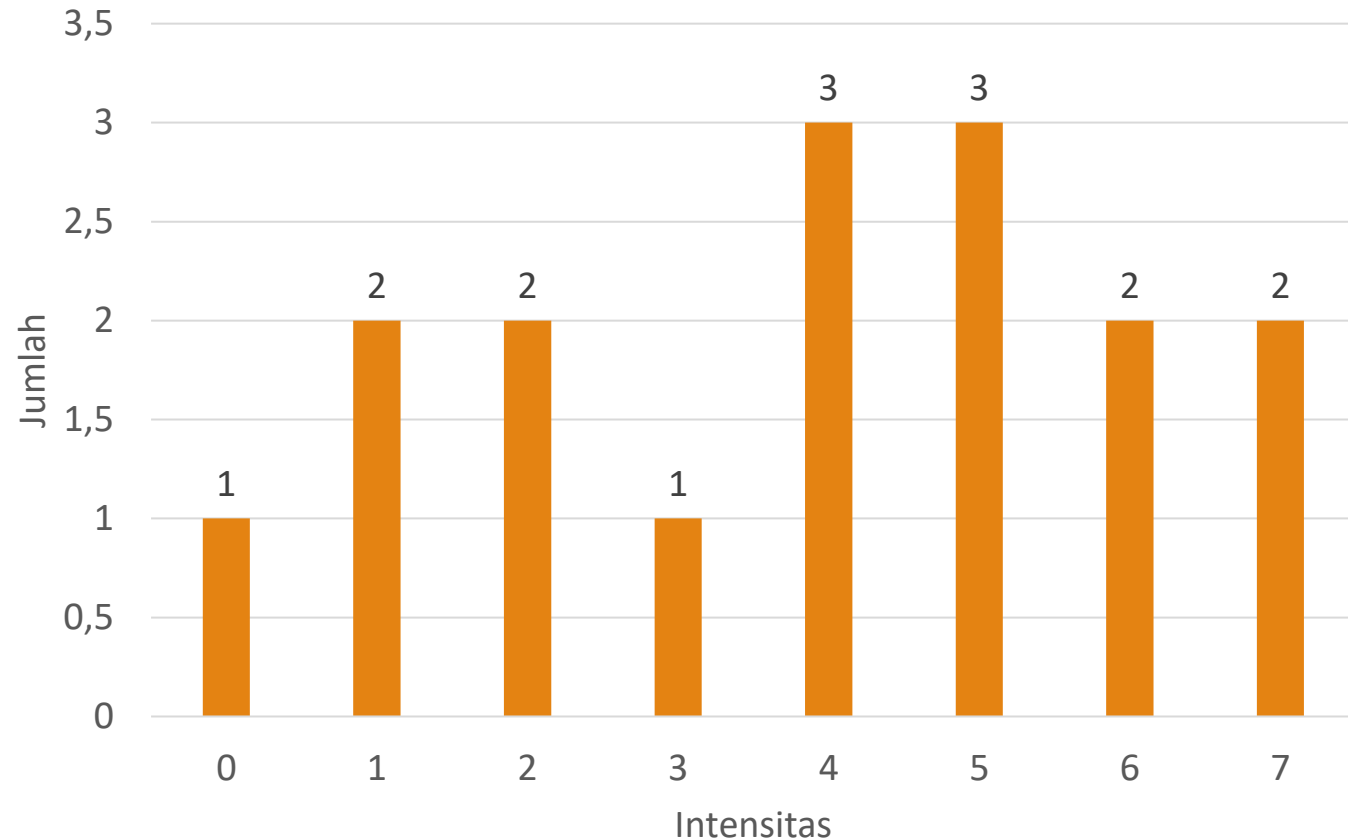
- Histogram menunjukkan frekuensi kemunculan dari nilai intensitas pada citra.
- Histogram dapat menunjukkan informasi tentang kecerahan (brightness) dan kontras (contrast) dari sebuah citra.
- Misal kita memiliki sebuah citra dengan rentang nilai keabuan atau intensitas $[0, L-1]$ dan kita ingin membuat histogramnya.
- Selanjutnya kita hitung frekuensi kemunculan nilai intensitas j dengan cara menghitung piksel yang memiliki nilai intensitas sebesar j .
- Langkah diatas digunakan untuk nilai $j = 0, 1, 2, \dots, L-1$.

Histogram

Menghitung Histogram

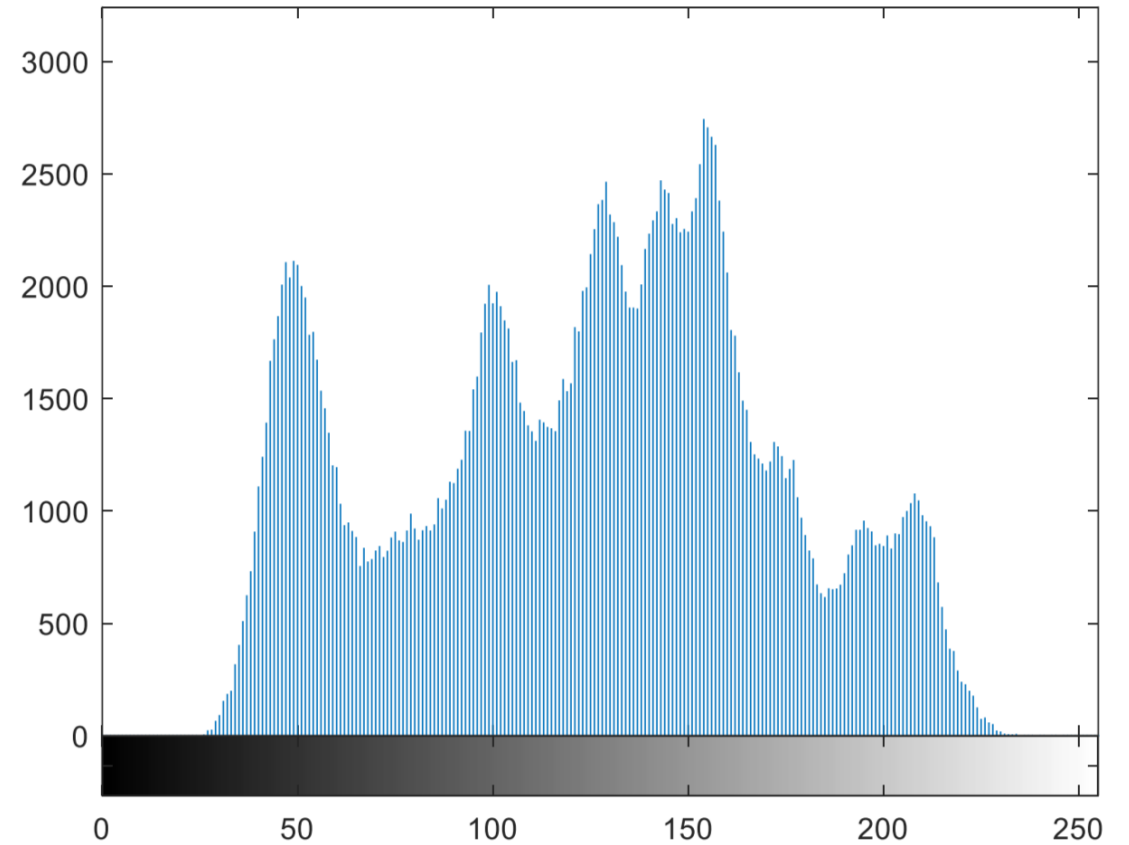
- Misal kita memiliki citra **3-bit** dengan rentang intensitas $[0, 7]$ ($2^3 - 1 = 7$) sebagai berikut, maka untuk mendapatkan histogram, kita akan menghitung jumlah tiap intensitasnya:

1	2	1	3
0	5	4	4
7	7	2	4
6	5	6	5



Histogram

Histogram Citra Grayscale (8-bit)



Histogram

Histogram Citra Grayscale (8-bit)

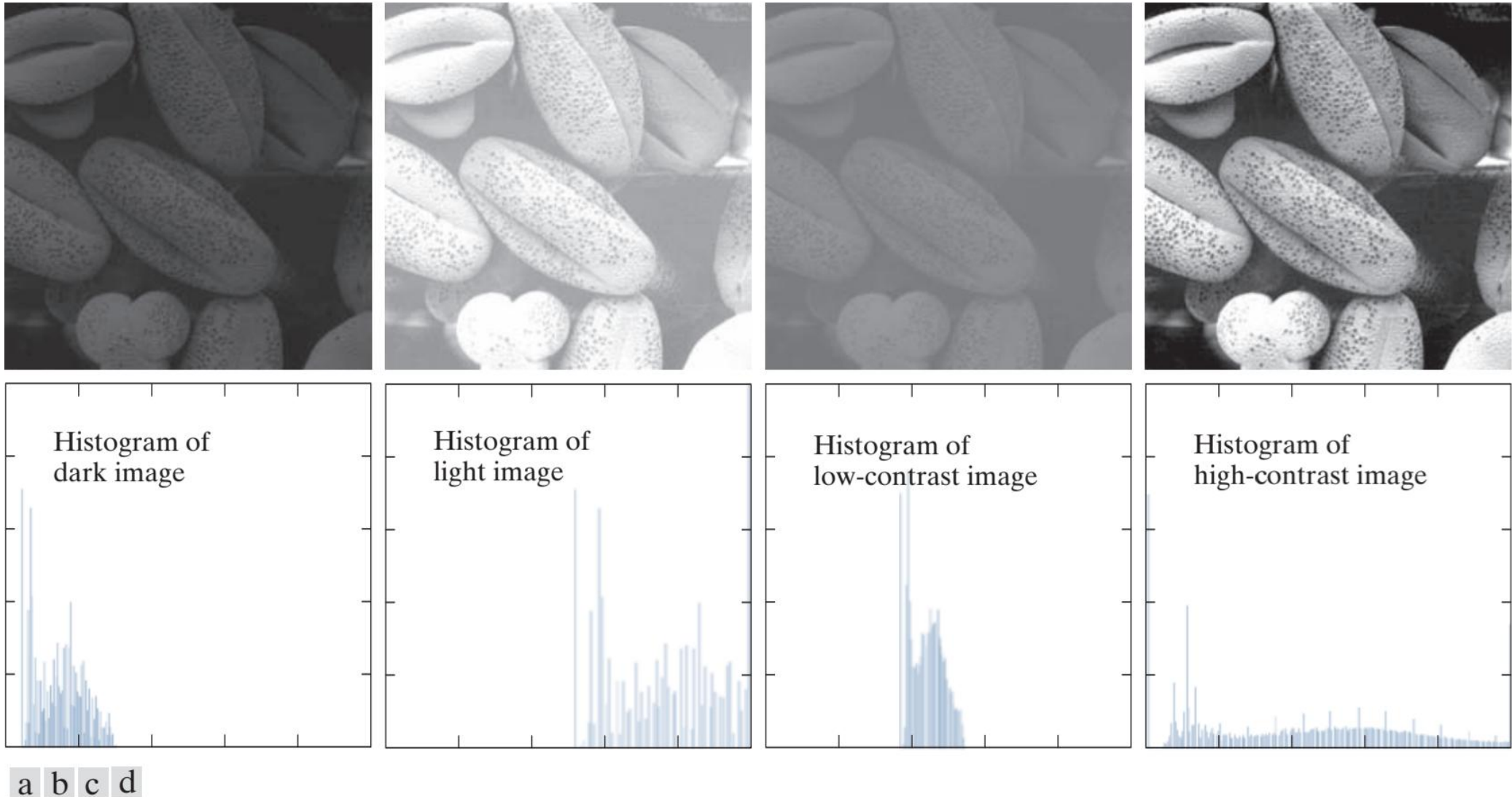
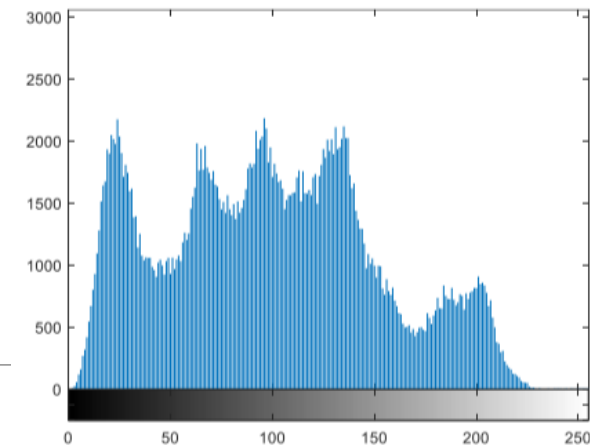


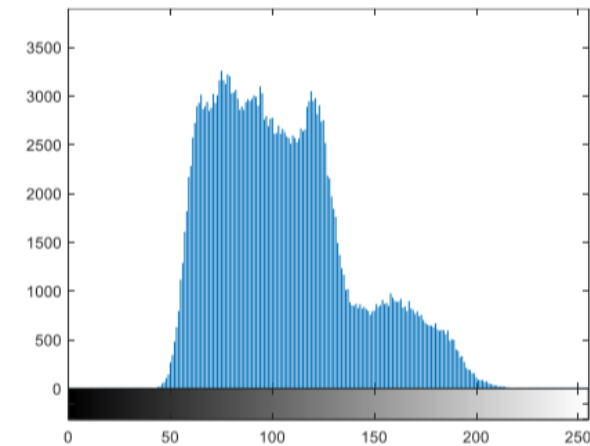
FIGURE 3.16 Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of r_k and the vertical axis are values of $p(r_k)$.

Histogram

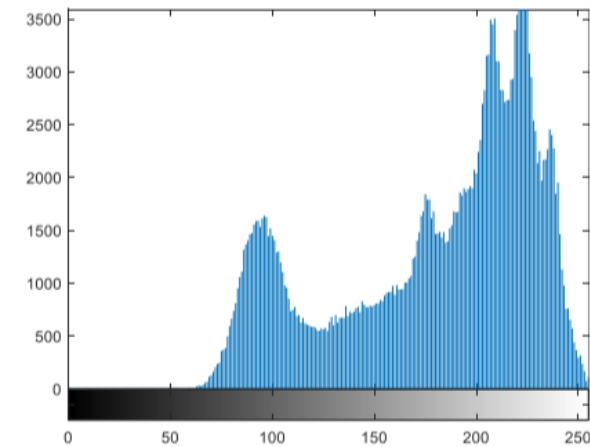
Histogram Citra RGB (24-bit)



Red



Green



Blue

Normalisasi Histogram

Histogram

Normalisasi Histogram

- The probability of occurrence of intensity level r_k in a digital image is approximated by:

$$p_r(r_k) = \frac{n_k}{MN} \quad (3-14)$$

- where M and N are the number of image rows and columns respectively, and n_k denotes the number of pixels that have intensity r_k .
- $p_r(r_k)$, with $r_k \in [0, L - 1]$, is commonly referred to as a normalized image histogram.

Histogram

Normalisasi Histogram

- Misalkan kita ingin melakukan normalisasi histogram dari citra **3-bit** pada [slide sebelumnya](#):

1	2	1	3
0	5	4	4
7	7	2	4
6	5	6	5

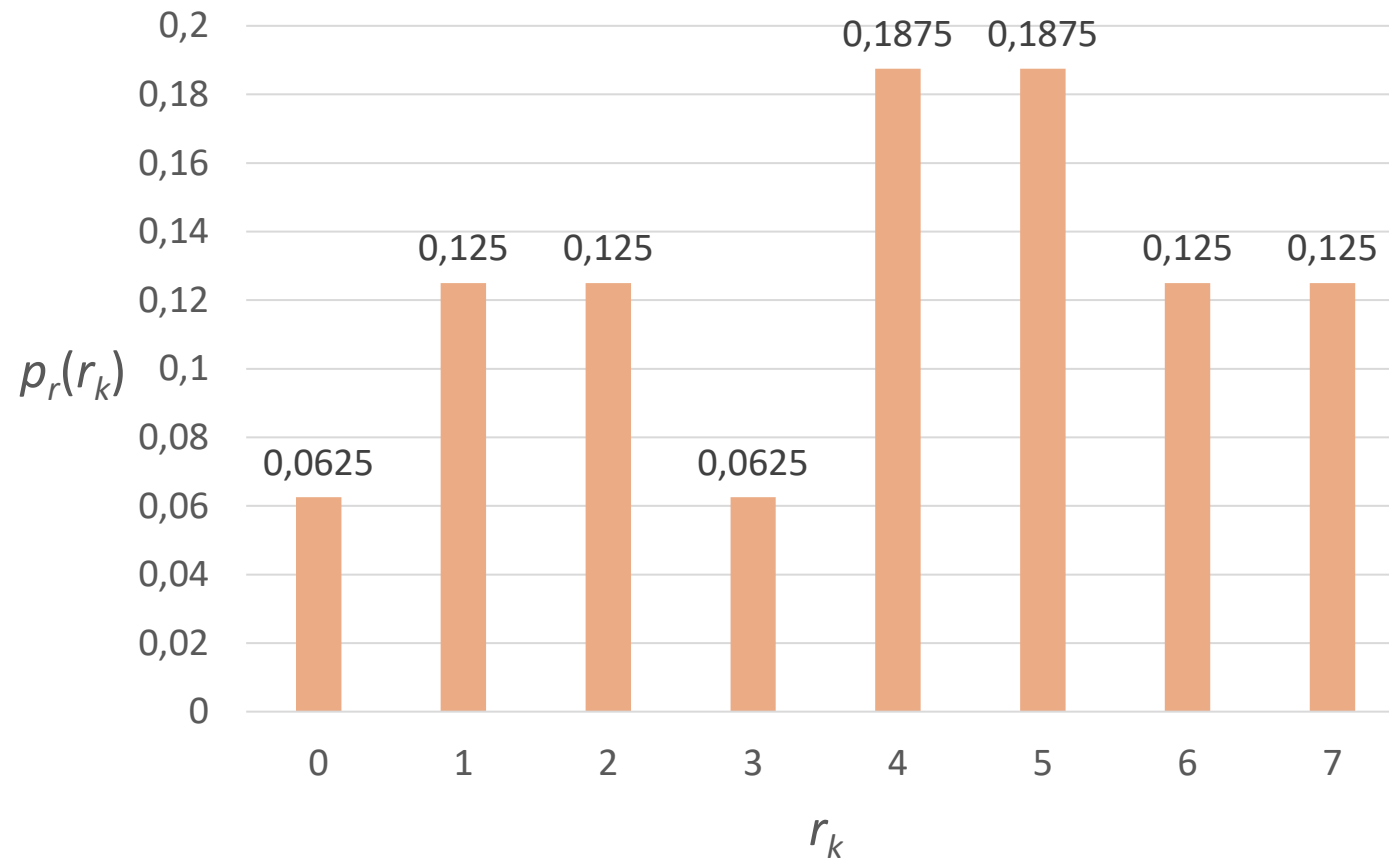
r_k	n_k	$p_r(r_k) = n_k / MN$ ($MN = 4 \times 4 = 16$)
$r_0 = 0$	1	0,0625
$r_1 = 1$	2	0,125
$r_2 = 2$	2	0,125
$r_3 = 3$	1	0,0625
$r_4 = 4$	3	0,1875
$r_5 = 5$	3	0,1875
$r_6 = 6$	2	0,125
$r_7 = 7$	2	0,125

- Jika semua nilai $p_r(r_k)$ dijumlahkan, maka hasilnya adalah 1.

Histogram

Normalisasi Histogram

- Sehingga didapatkan hasil normalisasi histogramnya:



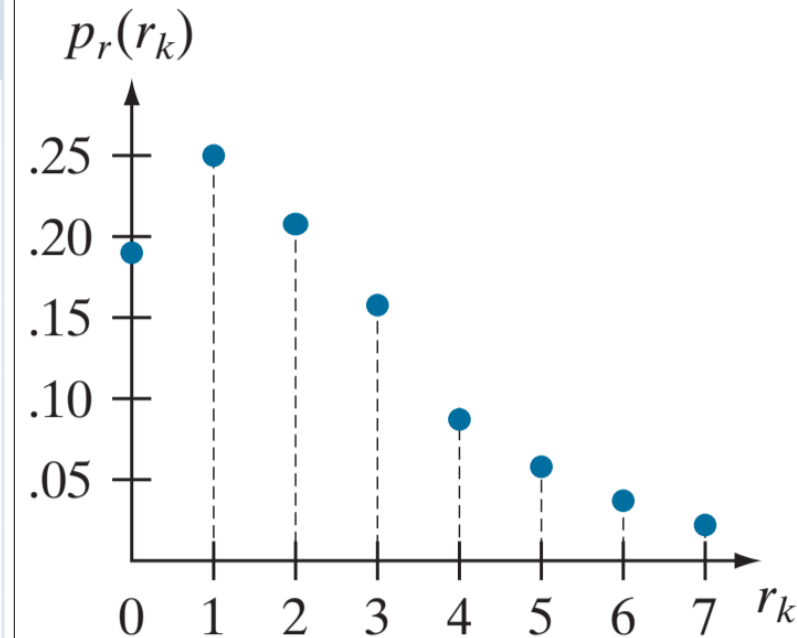
Histogram

Normalisasi Histogram

- Contoh lain dengan menggambarkan histogramnya; suppose that a 3-bit image ($L = 8$) of size 64×64 pixels ($MN = 64 \times 64 = 4096$) has the intensity distribution in Table 3.1, where the intensity levels are integers in the range $[0, L - 1] = [0, 8 - 1] = [0, 7]$.

TABLE 3.1
Intensity
distribution and
histogram values
for a 3-bit, 64×64
digital image.

r_k	n_k	$p_r(r_k) = n_k / MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02



Histogram Equalization

Histogram

Histogram Equalization

- **Histogram equalization:** mengubah nilai-nilai intensitas di dalam citra sehingga didapatkan **penyebarannya yang seragam (*uniform*)**.
- A processed (output) image is obtained by using Eq. (3-15) to map each pixel in the input image with intensity r_k into a corresponding pixel with level s_k in the output image. This is called a histogram equalization or histogram linearization transformation.

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j) \quad k = 0, 1, 2, \dots, L - 1 \quad (3-15)$$

Histogram

Histogram Equalization

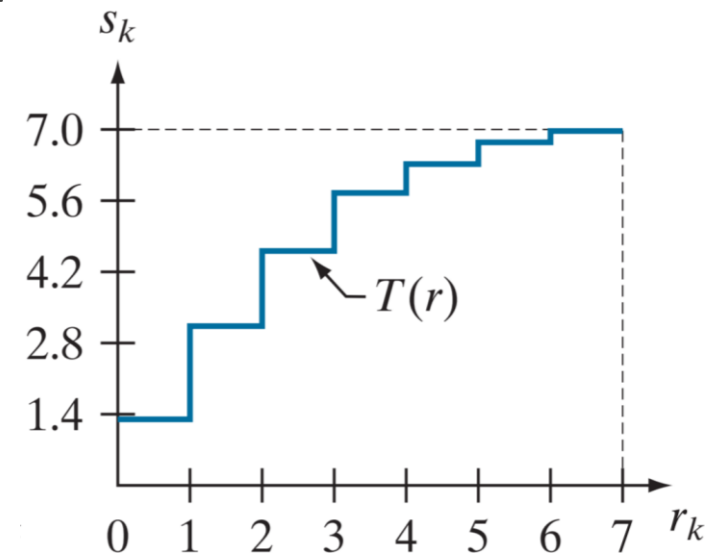
- Dari contoh [normalisasi histogram sebelumnya](#); values of the histogram equalization transformation function are obtained using Eq. (3-15). For instance,

$$s_0 = T(r_0) = (8 - 1) \sum_{j=0}^0 p_r(r_j) = 7p_r(r_0) = 7(0.19) = 1.33$$

$$s_1 = T(r_1) = (8 - 1) \sum_{j=0}^1 p_r(r_j) = 7(p_r(r_0) + p_r(r_1)) = 7(0.19 + 0.25) = 7(0.44) = 3.08$$

- Similarly, $s_1 = T(r_1) = 3.08$, $s_2 = 4.55$, $s_3 = 5.67$, $s_4 = 6.23$, $s_5 = 6.65$, $s_6 = 6.86$, and $s_7 = 7.00$. This transformation function has the staircase shape shown in:

r_k	n_k	$p_r(r_k) = n_k / MN$ ($MN = 16 \times 16 = 4096$)	s_k
$r_0 = 0$	790	0.19	1.33
$r_1 = 1$	1023	0.25	3.08
$r_2 = 2$	850	0.21	4.55
$r_3 = 3$	656	0.16	5.67
$r_4 = 4$	329	0.08	6.23
$r_5 = 5$	245	0.06	6.65
$r_6 = 6$	122	0.03	6.86
$r_7 = 7$	81	0.02	7.00



Histogram

Histogram Equalization

- At this point, the s values are fractional because they were generated by summing probability values, so we **round** them to their **nearest integer values** in the range $[0, 7]$:

$$\begin{array}{llll} s_0 = 1.33 \rightarrow 1 & s_2 = 4.55 \rightarrow 5 & s_4 = 6.23 \rightarrow 6 & s_6 = 6.86 \rightarrow 7 \\ s_1 = 3.08 \rightarrow 3 & s_3 = 5.67 \rightarrow 6 & s_5 = 6.65 \rightarrow 7 & s_7 = 7.00 \rightarrow 7 \end{array}$$

- These are the values of the equalized histogram. Observe that the transformation yielded only five distinct intensity levels (1, 3, 5, 6, 7).

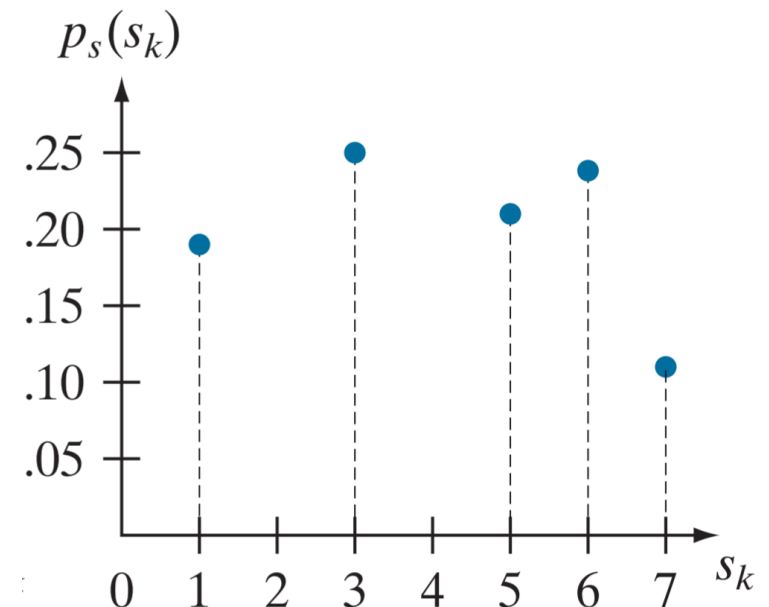
r_k	n_k	$p_r(r_k) = n_k / MN$ ($MN = 16 \times 16 = 4096$)	$s_k = \text{round}(s_k)$
$r_0 = 0$	790	0.19	1
$r_1 = 1$	1023	0.25	3
$r_2 = 2$	850	0.21	5
$r_3 = 3$	656	0.16	6
$r_4 = 4$	329	0.08	6
$r_5 = 5$	245	0.06	7
$r_6 = 6$	122	0.03	7
$r_7 = 7$	81	0.02	7

Histogram

Histogram Equalization

- Because $r_0 = 0$ was mapped to $s_0 = 1$, there are 790 pixels in the histogram equalized image with this value (see Table 3.1). Also, there are 1023 pixels with a value of $s_1 = 3$ and 850 pixels with a value of $s_2 = 5$. However, both r_3 and r_4 were mapped to the same value, 6, so there are $(656 + 329) = 985$ pixels in the equalized image with this value. Similarly, there are $(245 + 122 + 81) = 448$ pixels with a value of 7 in the histogram equalized image. Dividing these numbers by $MN = 4096$ yielded the equalized histogram in:

s_k	n_k	$p_s(s_k) = n_k / MN$ ($MN = 4096$)
1	790	$790 / 4096 = 0.19$
3	1023	$1023 / 4096 = 0.25$
5	850	$850 / 4096 = 0.21$
6	$656 + 329 = 985$	$985 / 4096 = 0.24$
7	$245 + 122 + 81 = 448$	$448 / 4096 = 0.11$



Histogram

Histogram Equalization

a b c

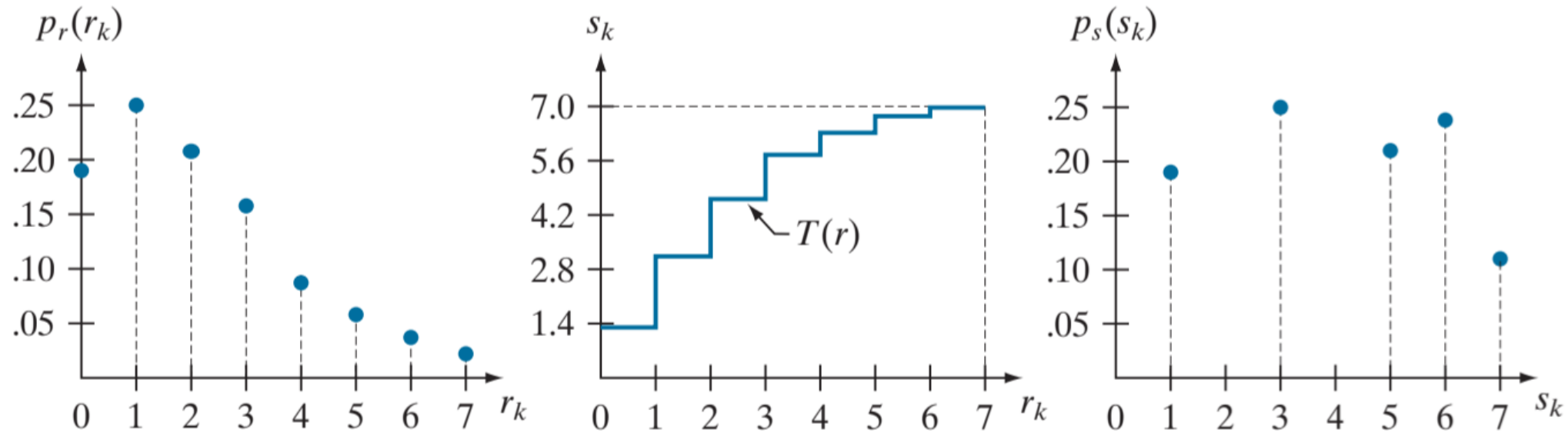
FIGURE 3.19

Histogram equalization.

(a) Original histogram.

(b) Transformation function.

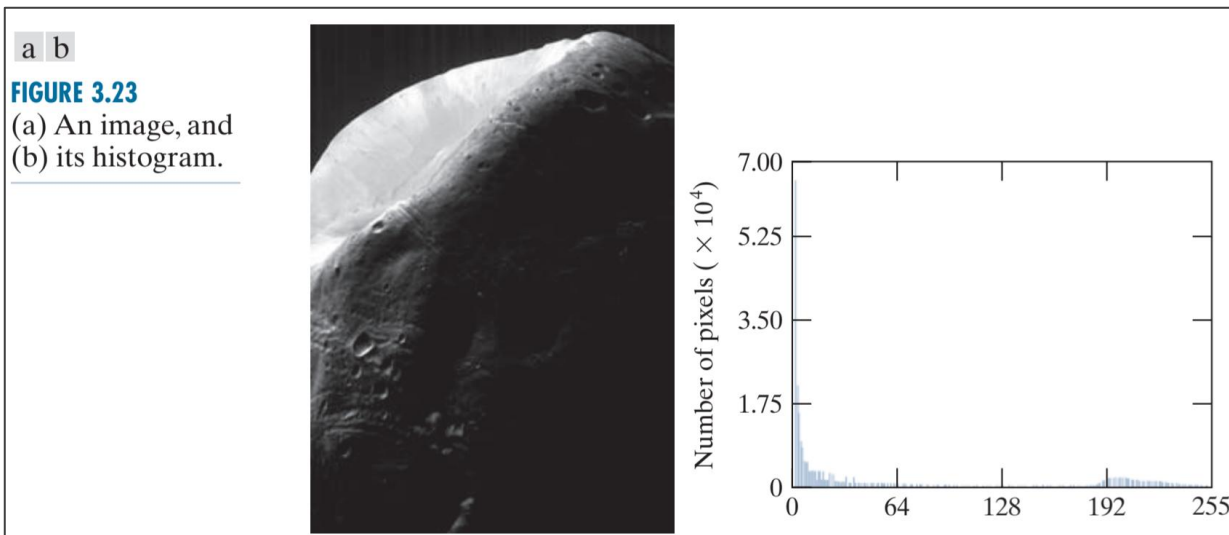
(c) Equalized histogram.



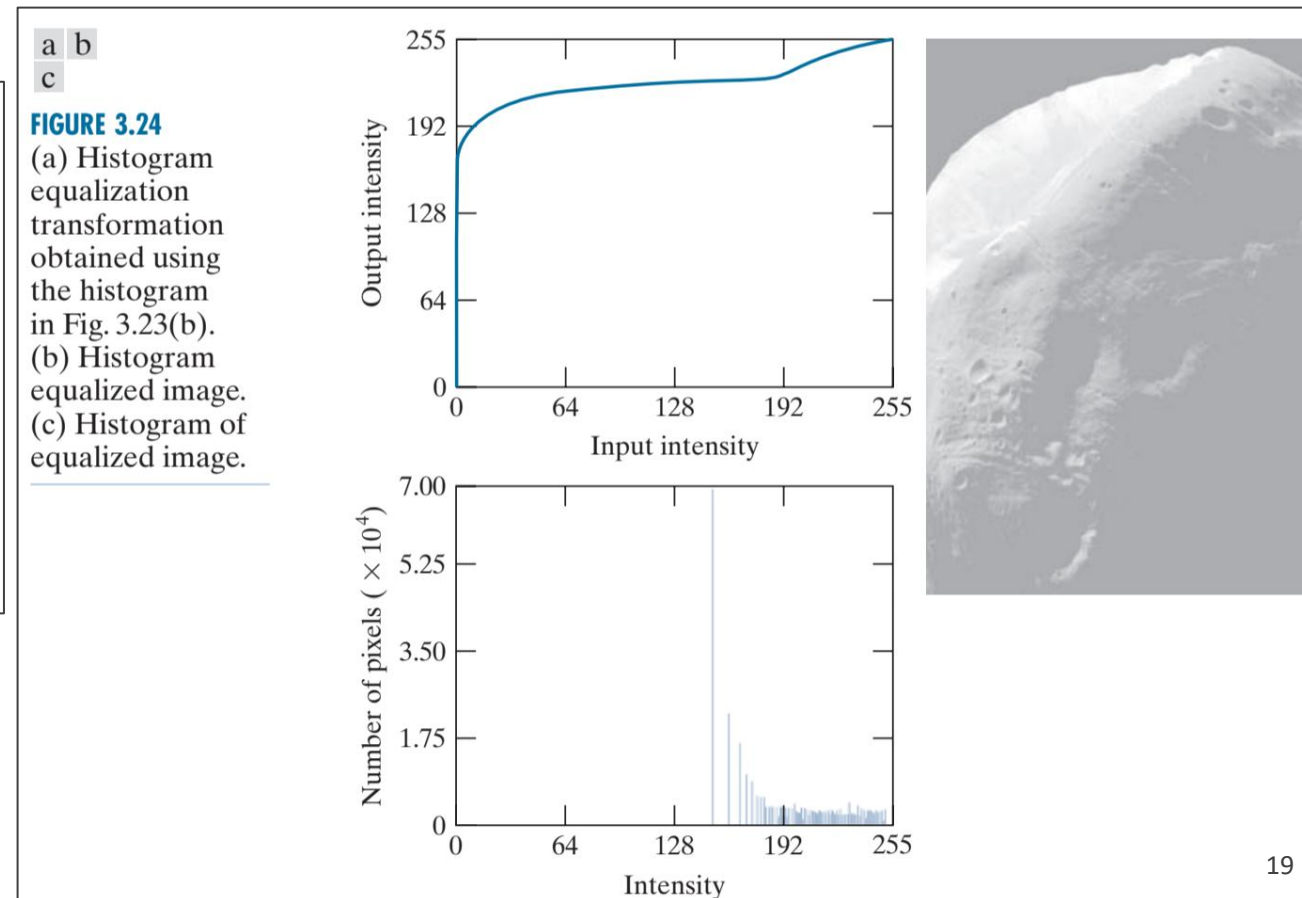
Histogram

Histogram Equalization

- Namun ternyata histogram equalization tidak selalu memberikan hasil yang baik. Pada contoh citra *Mars moon* berikut, hasil histogram equalization memberikan citra yang terlalu terang:



- Untuk mengatasi masalah tersebut, kita dapat menggunakan metode **histogram specification**.



Histogram Matching (Specification)

Histogram

Histogram Matching (Specification)

- **Histogram specification:** mengubah nilai-nilai intensitas di dalam citra supaya diperoleh histogram dengan bentuk yang **dispesifikasikan oleh pengguna**.
- Contoh tabel histogram sebelumnya, dan tabel histogram yang diinginkan (dispesifikasikan):

r_k	n_k	$p_r(r_k) = n_k / MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

Tabel histogram sebelumnya

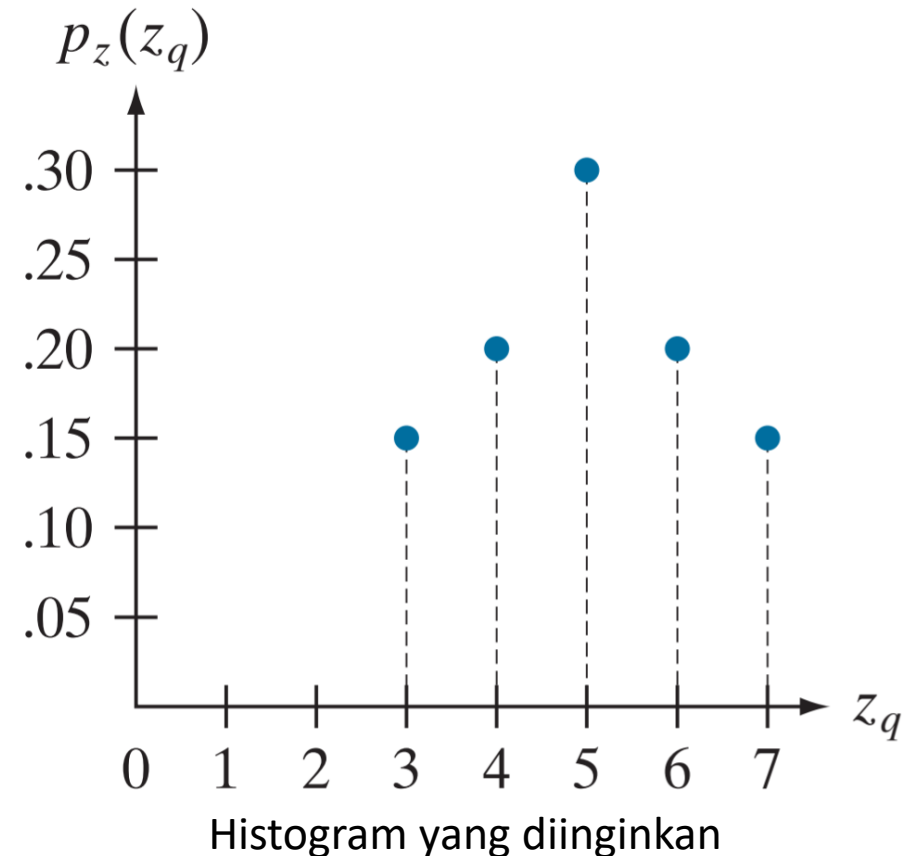
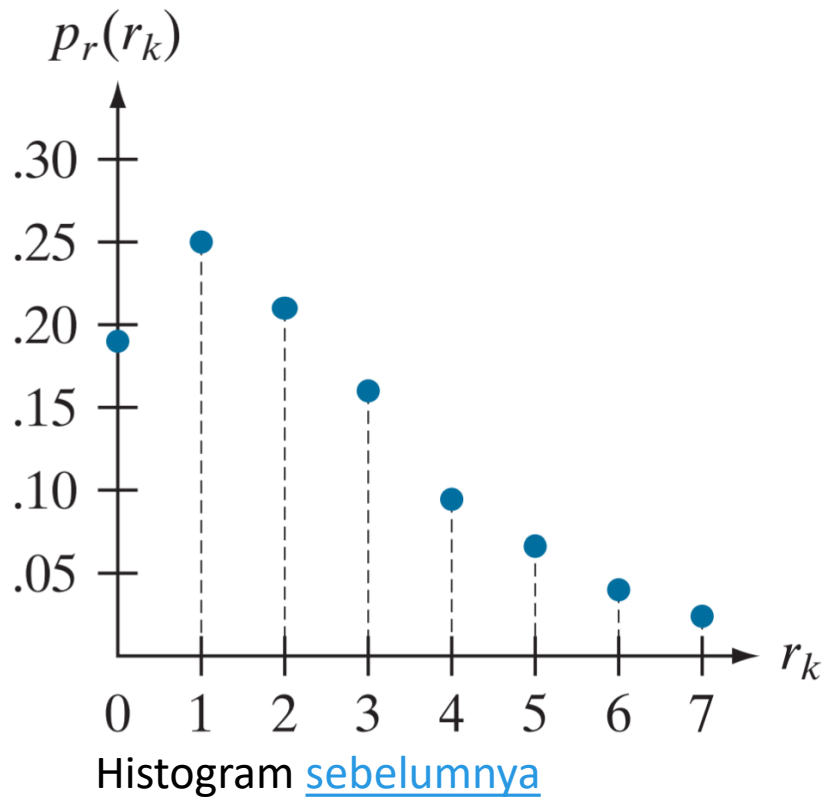
z_q	Specified $p_z(z_q)$
$z_0 = 0$	0.00
$z_1 = 1$	0.00
$z_2 = 2$	0.00
$z_3 = 3$	0.15
$z_4 = 4$	0.20
$z_5 = 5$	0.30
$z_6 = 6$	0.20
$z_7 = 7$	0.15

Tabel histogram yang diinginkan

Histogram

Histogram Matching (Specification)

- Dari tabel tadi, dapat digambarkan contoh histogram sebelumnya, dan histogram yang diinginkan (dispesifikasikan):



Histogram

Histogram Matching (Specification)

- Given a specific value of s_k , computing the transformation function:

$$G(z_q) = (L - 1) \sum_{i=0}^q p_z(z_i) \quad (3-21)$$

- Menghitung $G(z_q)$:

$$G(z_0) = (8 - 1) \sum_{i=0}^0 p_z(z_i) = 7(p_z(z_0)) = 7(0) = 0$$

$$G(z_1) = (8 - 1) \sum_{i=0}^1 p_z(z_i) = 7(p_z(z_0) + p_z(z_1)) = 7(0 + 0) = 0$$

$$G(z_2) = (8 - 1) \sum_{i=0}^2 p_z(z_i) = 7(p_z(z_0) + p_z(z_1) + p_z(z_2)) = 7(0 + 0 + 0) = 0$$

$$G(z_3) = (8 - 1) \sum_{i=0}^3 p_z(z_i) = 7(p_z(z_0) + p_z(z_1) + p_z(z_2) + p_z(z_3)) = 7(0 + 0 + 0 + 0.15) = 1.05$$

z_q	Specified $p_z(z_q)$
$z_0 = 0$	0.00
$z_1 = 1$	0.00
$z_2 = 2$	0.00
$z_3 = 3$	0.15
$z_4 = 4$	0.20
$z_5 = 5$	0.30
$z_6 = 6$	0.20
$z_7 = 7$	0.15

Tabel histogram yang diinginkan

Histogram

Histogram Matching (Specification)

- Didapatkan seluruh nilai $G(z_q)$ sebagai berikut:

$$G(z_0) = 0.00 \quad G(z_2) = 0.00 \quad G(z_4) = 2.45 \quad G(z_6) = 5.95$$

$$G(z_1) = 0.00 \quad G(z_3) = 1.05 \quad G(z_5) = 4.55 \quad G(z_7) = 7.00$$

- These fractional values are **rounded** to integers in the range $[0, 7]$:

$$G(z_0) = 0.00 \rightarrow 0 \quad G(z_4) = 2.45 \rightarrow 2$$

$$G(z_1) = 0.00 \rightarrow 0 \quad G(z_5) = 4.55 \rightarrow 5$$

$$G(z_2) = 0.00 \rightarrow 0 \quad G(z_6) = 5.95 \rightarrow 6$$

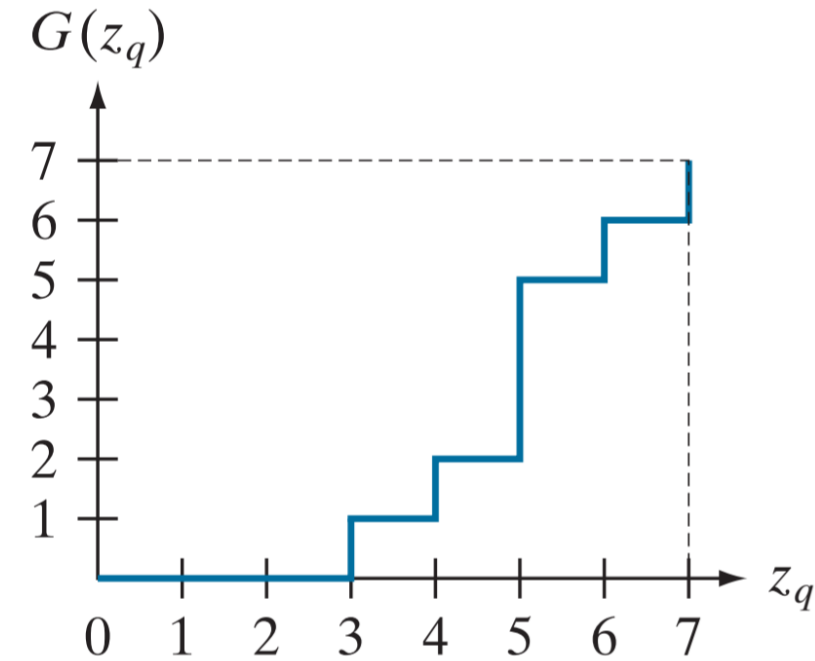
$$G(z_3) = 1.05 \rightarrow 1 \quad G(z_7) = 7.00 \rightarrow 7$$

Histogram

Histogram Matching (Specification)

TABLE 3.3
Rounded values
of the
transformation
function $G(z_q)$.

z_q	$G(z_q)$
$z_0 = 0$	0
$z_1 = 1$	0
$z_2 = 2$	0
$z_3 = 3$	1
$z_4 = 4$	2
$z_5 = 5$	5
$z_6 = 6$	6
$z_7 = 7$	7



Histogram

Histogram Matching (Specification)

- Next, find the **smallest value of z_q** so that the **value $G(z_q)$ is the closest to s_k** .
- When more than one value of z_q gives the same match (i.e., the mapping is not unique), choose the **smallest value** by convention.

Histogram

Histogram Matching (Specification)

- Didapatkan:

- $s_k = 1$ paling dekat dengan $1 = G(z_3)$
- $s_k = 3$ paling dekat dengan $2 = G(z_4)$
- $s_k = 5$ paling dekat dengan $5 = G(z_5)$
- $s_k = 6$ paling dekat dengan $6 = G(z_6)$
- $s_k = 7$ paling dekat dengan $7 = G(z_7)$

$s_k = 3$ berada di antara $G(z_4) = 2$ dan $G(z_5) = 5$ (3 berada di antara 2 dan 5), maka 3 lebih dekat dengan 2 (selisih 1) daripada 5 (selisih 2).

TABLE 3.3

Rounded values of the transformation function $G(z_q)$.

z_q	$G(z_q)$
$z_0 = 0$	0
$z_1 = 1$	0
$z_2 = 2$	0
$z_3 = 3$	1
$z_4 = 4$	2
$z_5 = 5$	5
$z_6 = 6$	6
$z_7 = 7$	7

s_k	n_k	$p_s(s_k) = n_k / MN$ ($MN = 4096$)
1	790	$790 / 4096 = 0.19$
3	1023	$1023 / 4096 = 0.25$
5	850	$850 / 4096 = 0.21$
6	$656 + 329 = 985$	$985 / 4096 = 0.24$
7	$245 + 122 + 81 = 448$	$448 / 4096 = 0.11$

Hasil histogram equalization [sebelumnya](#)

Histogram

Histogram Matching (Specification)

- In the final step of the procedure, we use the mappings in Table 3.4 to map every pixel in the histogram equalized image into a corresponding pixel in the newly created histogram-specified image.

TABLE 3.4

Mapping of values s_k into corresponding values z_q .

s_k	$\rightarrow$	z_q
1	$\rightarrow$	3
3	$\rightarrow$	4
5	$\rightarrow$	5
6	$\rightarrow$	6
7	$\rightarrow$	7

Histogram

Histogram Matching (Specification)

- We will find the value of $p_z(z_q)$. For instance, we see in Table 3.4 that $s_k = 1$ maps to $z_q = 3$, and there are 790 pixels in the [histogram equalized](#) image with a value of 1. Therefore, $p_z(z_3) = 790/4096 = 0.19$.
- Jadi, hasil $p_z(z_q)$ juga dapat diambil dari $p_s(s_k)$ pada [histogram equalization](#) sebelumnya.

TABLE 3.3 Rounded values of the transformation function $G(z_q)$.	z_q	$G(z_q)$	s_k	n_k	$p_s(s_k) = n_k / MN$ ($MN = 4096$)	
	$z_0 = 0$	0				
	$z_1 = 1$	0	1	790	$790 / 4096 = 0.19$	$\leq p_z(z_3)$
	$z_2 = 2$	0	3	1023	$1023 / 4096 = 0.25$	$\leq p_z(z_4)$
	$z_3 = 3$	1	5	850	$850 / 4096 = 0.21$	$\leq p_z(z_5)$
	$z_4 = 4$	2	6	$656 + 329 = 985$	$985 / 4096 = 0.24$	$\leq p_z(z_6)$
	$z_5 = 5$	5	7	$245 + 122 + 81 = 448$	$448 / 4096 = 0.11$	$\leq p_z(z_7)$
	$z_6 = 6$	6				
	$z_7 = 7$	7				

Hasil histogram equalization [sebelumnya](#)

Histogram

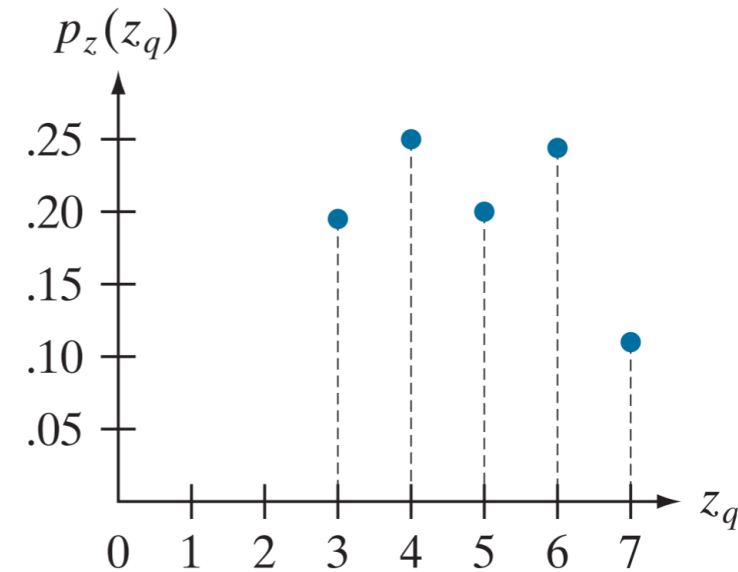
Histogram Matching (Specification)

- Setelah mendapatkan nilai dari $p_z(z_q)$, maka hasil histogram matching / specification (**actual**):

TABLE 3.2

Specified and actual histograms (the values in the third column are computed in Example 3.7).

z_q	Specified $p_z(z_q)$	Actual $p_z(z_q)$
$z_0 = 0$	0.00	0.00
$z_1 = 1$	0.00	0.00
$z_2 = 2$	0.00	0.00
$z_3 = 3$	0.15	0.19 $\leq p_z(z_3)$
$z_4 = 4$	0.20	0.25 $\leq p_z(z_4)$
$z_5 = 5$	0.30	0.21 $\leq p_z(z_5)$
$z_6 = 6$	0.20	0.24 $\leq p_z(z_6)$
$z_7 = 7$	0.15	0.11 $\leq p_z(z_7)$



Histogram

Histogram Matching (Specification)

a	b
c	d

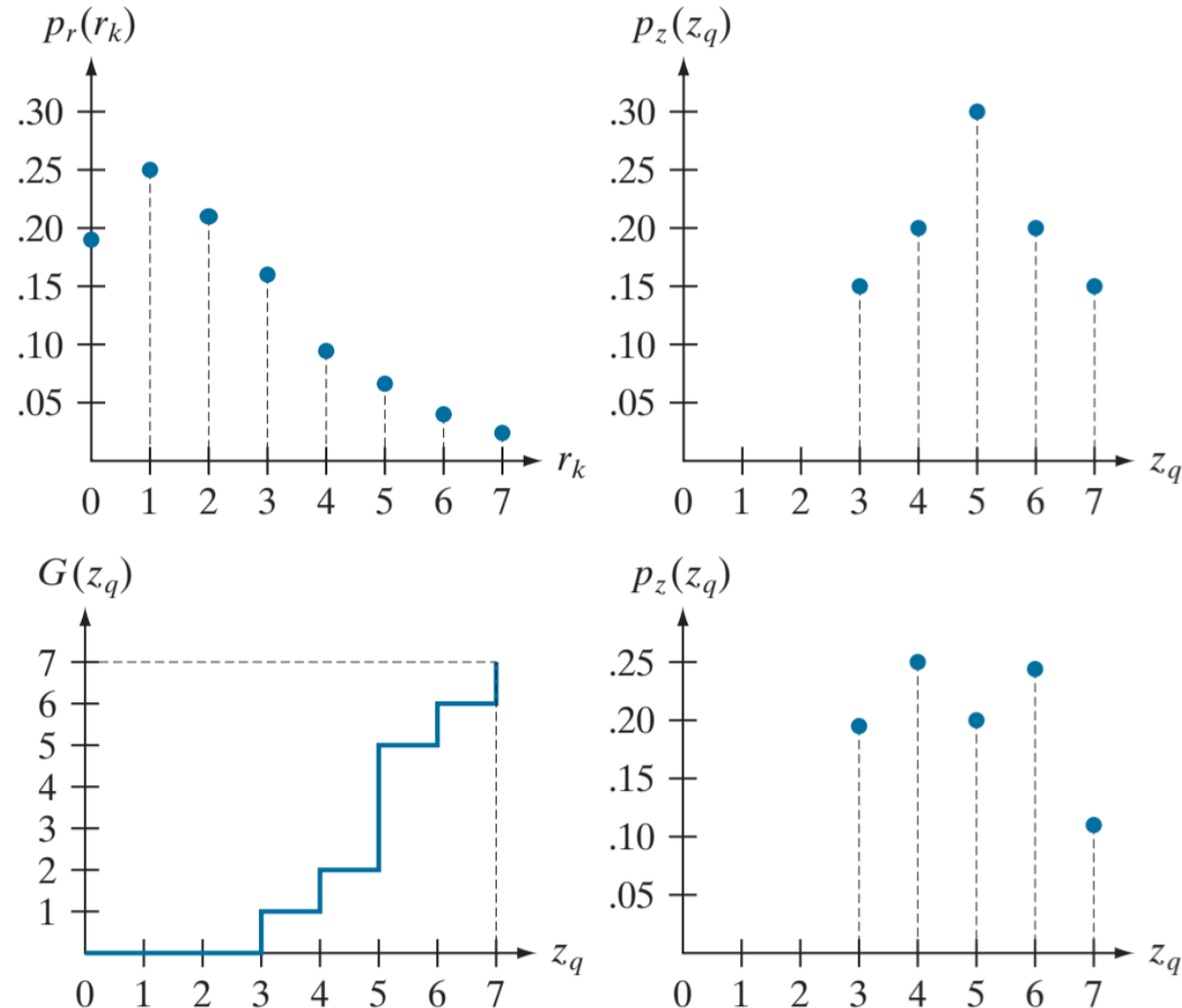
FIGURE 3.22

(a) Histogram of a 3-bit image.

(b) Specified histogram.

(c) Transformation function obtained from the specified histogram.

(d) Result of histogram specification. Compare the histograms in (b) and (d).



Histogram Histogram Matching (Specification)

a c
b
d

FIGURE 3.25
Histogram specification.
(a) Specified histogram.
(b) Transformation $G(z_q)$, labeled (1), and $G^{-1}(s_k)$, labeled (2).
(c) Result of histogram specification.
(d) Histogram of image (c).

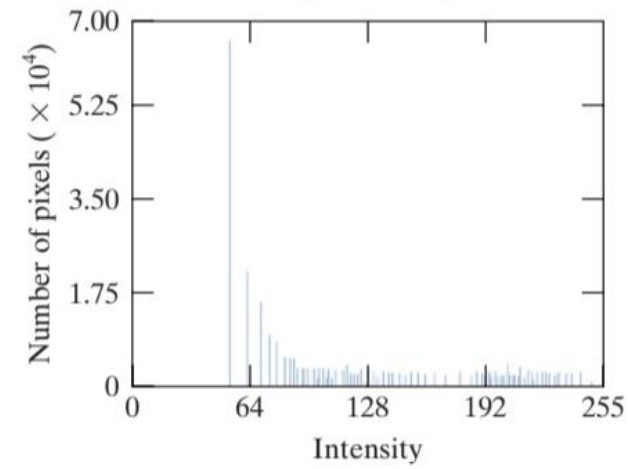
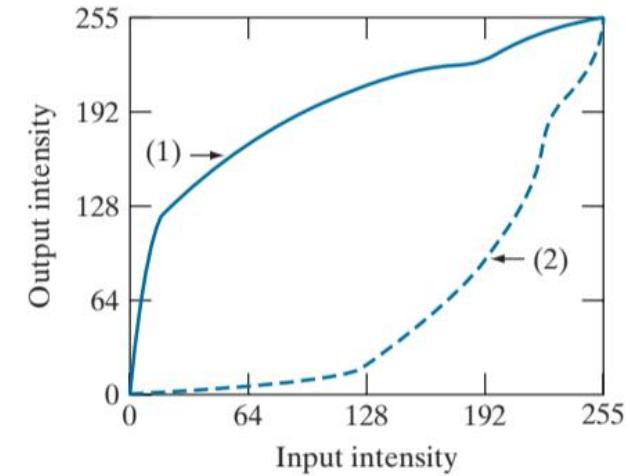
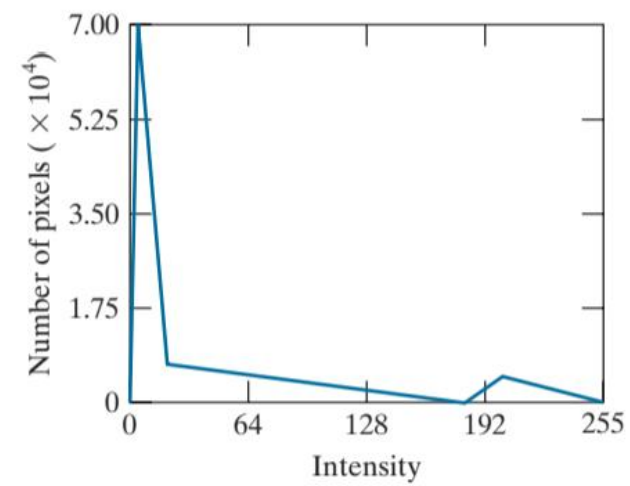


Image Enhancement dengan Histogram

Image Enhancement dengan Histogram

Perbaikan Lokal dengan Histogram Equalization

- Histogram equalization tidak hanya dilakukan untuk seluruh piksel pada citra, tetapi dapat juga hanya untuk area tertentu pada citra.
- Ini berarti dapat dilakukan **perbaikan secara lokal** dengan metode histogram equalization.

Image Enhancement dengan Histogram

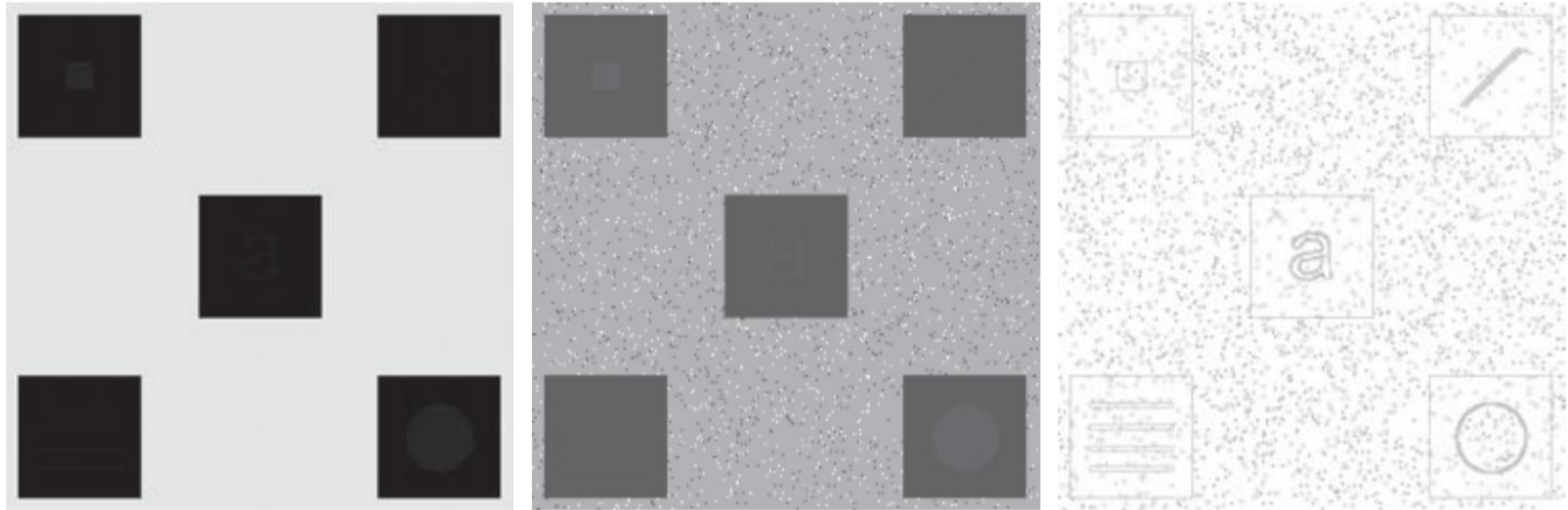
Perbaikan Lokal dengan Histogram Equalization

- Figure 3.26(c) was obtained using local histogram equalization of Fig. 3.26(a) with a neighborhood of size 3×3 .

a b c

FIGURE 3.26

(a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization.



Terima Kasih
